

Supplementary Material

MTGDriver: Multitask Graph Learning for Cancer Driver Gene Prioritization

Table S1. Comparison of MTGDriver with baseline methods across 12 cancer types, evaluated using AUPRC (mean $\pm$ standard deviation).

Cancer	MTGDriver	EMOGI	MTGCN	DGHNN
BRCA	0.7306 $\pm$ 0.0096	0.6482 $\pm$ 0.0045	0.6583 $\pm$ 0.0040	0.6832 $\pm$ 0.0062
BLCA	0.7155 $\pm$ 0.0219	0.5485 $\pm$ 0.0101	0.6568 $\pm$ 0.0077	0.6935 $\pm$ 0.0064
LUAD	0.6653 $\pm$ 0.0096	0.5591 $\pm$ 0.0105	0.6279 $\pm$ 0.0099	0.6410 $\pm$ 0.0091
LIHC	0.5076 $\pm$ 0.0196	0.3845 $\pm$ 0.0193	0.4645 $\pm$ 0.0195	0.4811 $\pm$ 0.0191
THCA	0.3533 $\pm$ 0.0153	0.2587 $\pm$ 0.0084	0.2907 $\pm$ 0.0110	0.3145 $\pm$ 0.0233
LUSC	0.4461 $\pm$ 0.0536	0.2222 $\pm$ 0.0266	0.3099 $\pm$ 0.0407	0.4054 $\pm$ 0.0542
ESCA	0.5027 $\pm$ 0.0289	0.4174 $\pm$ 0.0297	0.4721 $\pm$ 0.0279	0.4763 $\pm$ 0.0145
PRAD	0.7468 $\pm$ 0.0256	0.5233 $\pm$ 0.0220	0.6143 $\pm$ 0.0189	0.7058 $\pm$ 0.0129
STAD	0.7240 $\pm$ 0.0168	0.4604 $\pm$ 0.0153	0.5931 $\pm$ 0.0117	0.6449 $\pm$ 0.0083
COAD	0.5353 $\pm$ 0.0205	0.3459 $\pm$ 0.0067	0.3816 $\pm$ 0.0085	0.4468 $\pm$ 0.0220
UCEC	0.7121 $\pm$ 0.0235	0.4020 $\pm$ 0.0143	0.4954 $\pm$ 0.0145	0.5342 $\pm$ 0.0250
CESC	0.5715 $\pm$ 0.0687	0.5547 $\pm$ 0.0522	0.5972 $\pm$ 0.0528	0.5635 $\pm$ 0.0414

Table S2. Performance of Single-task and Multi-task settings across 12 cancer types, evaluated using AUPRC (mean $\pm$ standard deviation).

Cancer	Single-task	Multi-task
BRCA	0.6233 $\pm$ 0.0100	0.7110 $\pm$ 0.0125
BLCA	0.7104 $\pm$ 0.0114	0.7028 $\pm$ 0.0191
LUAD	0.6473 $\pm$ 0.0052	0.6508 $\pm$ 0.0157
LIHC	0.4556 $\pm$ 0.0209	0.4863 $\pm$ 0.0226
THCA	0.2435 $\pm$ 0.0144	0.3107 $\pm$ 0.0309
LUSC	0.3054 $\pm$ 0.0377	0.4197 $\pm$ 0.0402
ESCA	0.4000 $\pm$ 0.0429	0.4926 $\pm$ 0.0186
PRAD	0.6577 $\pm$ 0.0390	0.7021 $\pm$ 0.0315
STAD	0.6796 $\pm$ 0.0261	0.6938 $\pm$ 0.0235
COAD	0.3978 $\pm$ 0.0237	0.5136 $\pm$ 0.0212
UCEC	0.5750 $\pm$ 0.0190	0.6837 $\pm$ 0.0238
CESC	0.5098 $\pm$ 0.0521	0.4628 $\pm$ 0.0281

Table S3. Comparison of MTGDriver without and with pretraining across 12 cancer types, evaluated using AUPRC (mean $\pm$ standard deviation).

Cancer	W/O Pretraining	Pretraining
BRCA	0.7110 $\pm$ 0.0125	0.7306 $\pm$ 0.0096
BLCA	0.7028 $\pm$ 0.0191	0.7155 $\pm$ 0.0219
LUAD	0.6508 $\pm$ 0.0157	0.6653 $\pm$ 0.0096
LIHC	0.4863 $\pm$ 0.0226	0.5076 $\pm$ 0.0196
THCA	0.3107 $\pm$ 0.0309	0.3533 $\pm$ 0.0153
LUSC	0.4197 $\pm$ 0.0402	0.4461 $\pm$ 0.0536
ESCA	0.4926 $\pm$ 0.0186	0.5027 $\pm$ 0.0289
PRAD	0.7021 $\pm$ 0.0315	0.7468 $\pm$ 0.0256
STAD	0.6938 $\pm$ 0.0235	0.7240 $\pm$ 0.0168
COAD	0.5136 $\pm$ 0.0212	0.5353 $\pm$ 0.0205
UCEC	0.6837 $\pm$ 0.0238	0.7121 $\pm$ 0.0235
CESC	0.4628 $\pm$ 0.0281	0.5715 $\pm$ 0.0687

Table S4. Summary statistics of feature counts and gene counts across 12 cancer types.

	BRCA	BLCA	LUAD	THCA	LIHC	LUSC	ESCA	PRAD	STAD	COAD	UCEC	CESC
Mutation-derived features	4	4	4	4	4	4	4	4	4	4	4	4
Pathway-derived features	8	6	7	5	11	7	5	8	8	9	8	14
Total features	12	10	11	9	15	11	9	12	12	13	12	18
Driver genes	202	95	179	64	82	26	71	50	73	141	68	17
Passenger genes	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187

Table S5. Over-Representation Analysis (ORA) for BRCA and PRAD using representative cancer-related HDO and KEGG terms.

Cancer	Collection	Top-K	Count	Fold enrichment	FDR (BH)
BRCA	HDO	100	17	3.38	1.55×10^{-4}
		150	23	3.01	3.88×10^{-5}
		200	27	2.73	3.25×10^{-5}
	KEGG	100	8	6.14	1.30×10^{-4}
		150	11	5.41	1.25×10^{-5}
		200	16	5.99	2.89×10^{-8}
PRAD	HDO	50	13	6.09	2.70×10^{-5}
		100	26	6.28	5.47×10^{-12}
		150	37	6.07	2.17×10^{-12}
		200	47	5.81	1.61×10^{-21}
	KEGG	50	15	28.42	9.52×10^{-17}
		100	30	27.37	4.80×10^{-35}
		150	32	19.71	2.77×10^{-32}
		200	39	17.57	3.01×10^{-38}